MetroCity

Smart City Traffic & Accident Analytics

A Complete Story: Raw CSVs → Python ETL → MySQL Star Schema → Power BI

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1. Project Objectives & Business Context

1.1 Problem Statement

Welcome to **MetroCity**, a rapidly growing urban hub facing severe traffic congestion and frequent road accidents. The city administration wants to build a **Smart Traffic Management System** to reduce congestion and improve road safety using data analytics.

Despite installing 300 traffic sensors across five city zones, the administration cannot turn raw readings and accident records into decisions. Three core problems drive this project:

- **No congestion visibility:** Sensor data records a congestion label (Low/Moderate/High) but no numeric index; decisions cannot be ranked or forecasted.
- **No accident risk ranking:** 5,000 accident records exist but are unanalysed; the city cannot identify which locations, weather conditions, or vehicle types produce the worst outcomes.

- **No predictive capability:** No model exists to flag high-risk conditions before accidents occur, limiting pre-emptive emergency positioning and dynamic signage.

1.2 Business Goals

- **Analyse traffic flow** across five zones and all 24 hours to map congestion hotspots and identify peak windows.
- **Rank accident-prone locations** using a composite Risk Score (severity × log1p(casualties+1) × vehicles) and identify weather and road-condition triggers.
- **Build a predictive ML model** to classify whether a given accident scenario will produce a Fatal or Severe outcome.
- **Deliver a Power BI dashboard** across five thematic pages telling a complete road-safety story to the MetroCity administration.

1.3 Technology Stack

Technology	Role in Pipeline
Python (pandas, numpy, scikit-learn, mysql-connector-python)	Full ETL pipeline: extract, transform, feature-engineer, load. ML model training and risk probability scoring.
MySQL (smart_city_dw)	Star schema data warehouse: 7 tables + 8 pre-aggregated views. All views are Power BI-ready pre-joins.
Power BI (DAX + Power Query)	5 dashboard pages, 7 active relationships, 6 DAX measures, conditional formatting for all KPI columns.
scikit-learn	3 models trained & compared (Random Forest, Gradient Boosting, Logistic Regression) via 5-fold stratified CV. Best model selected by ROC-AUC.
SQL (analytical views)	9 validation queries + 8 MySQL views that cross-validate all Python EDA findings and serve as Power BI data sources.

1.4 Key Performance Indicators (KPIs)

KPI #	KPI Name	Definition & Value
1	Total Accidents	Count of all accident records: 5,000
2	Total Casualties	Sum of all casualty figures: 22,467
3	Fatal Accidents	Accident_Severity = Fatal: 1,253 (25.1%)
4	Severe Accidents	Accident_Severity = Severe: 1,237 (24.7%)
5	Fatal or Severe %	(Fatal + Severe) / Total: 49.8%
6	Avg Risk Score	Mean of severity × log1p(casualties+1) × vehicles: 11.04
7	Max Risk Score	Worst single accident composite score: 38.37
8	Avg Casualties/Accident	Mean casualties per incident: 4.49
9	Adverse Weather %	Accidents under Fog / Snow / Storm: 58.5%
10	Hazardous Road %	Accidents on Icy / Under Construction roads: 50.2%
11	Peak Hour Accident %	Accidents during 07-10h & 17-20h: 33.3%
12	Total Sensor Readings	Traffic sensor records processed: 300

13	Total Vehicles Recorded	Sum of Vehicle_Count across all sensor readings: 83,623
14	Avg Congestion Index	Mean of engineered 0-1 composite index: 0.490
15	High Congestion %	Sensor readings classified High: 32.0%
16*	Accident Rate per 1,000 Vehicles	Standard traffic safety KPI: accidents / total vehicles × 1,000. Engineered in vw_location_summary.
17*	ML Predicted Risk Probability	Per-accident probability of Fatal/Severe from best ML model (Random Forest).
18*	Risk Tier	Actionable 4-tier bucket: Low / Moderate / High / Critical Risk based on predicted probability.

2. Data Sources & Initial Assessment

2.1 Source Files

File	Rows	Columns
road_traffic_sensor_data.csv	300	Sensor_ID, Location, Date_Time, Vehicle_Count, Average_Speed, Congestion_Level
traffic_accident_data.csv	5,000	Accident_ID, Date_Time, Location, Weather_Condition, Road_Condition, Vehicle_Type, Accident_Severity, Number_of_Vehicles, Casualties, Traffic_Density

2.2 Data Quality Assessment

Both files were completely clean; zero nulls, zero duplicate primary keys, no negative values. Three structural gaps required engineering rather than cleaning:

- **No derived KPI columns:** No congestion index, risk score, severity score, or hazard classification exists in either raw file. All analytical metrics were engineered in the Transform stage.
- **No temporal decomposition:** Date_Time was a raw string. Hour, day_of_week, month, quarter, time_of_day, is_peak_hour, and is_weekend were all extracted during Transform.
- **No vehicle classification hierarchy:** Vehicle_Type contained five raw values with no grouping. A vehicle_class column (Heavy Motor / Light Motor / Two-Wheeler / Non-Motorised) was created for Power BI hierarchy slicing.

Assumption: All Sensors Are Active

The raw sensor file contains no status column. Per the project specification, all traffic sensor statuses are Active. The string "Active" was applied uniformly to the sensor_status column in fact_traffic.

Assumption: Zero-Casualty Fatal Accidents : log1p(Casualties + 1)

501 accidents have Casualties = 0 in the raw data, including 120 records labelled Fatal severity.

If the Risk Score used log1p(Casualties) directly, then log1p(0) = 0 and any zero-casualty Fatal accident would score zero; identical to a Minor accident with zero casualties. This loses the severity signal entirely.

Solution: The formula uses log1p(Casualties + 1), giving log1p(0+1) = 0.693 as the minimum floor. This ensures every Fatal accident scores higher than zero regardless of the casualty count recorded.

3. Python ETL Pipeline

The pipeline follows a strict Extract → Transform → Load sequence. A reusable get_conn() helper manages all MySQL connections; exec_sql() handles single and bulk inserts with error propagation; qdf() executes any SELECT and returns a pandas DataFrame.

3.1 Extract

```
raw_sensor   = pd.read_csv(SENSOR_FILE)
raw_accident = pd.read_csv(ACCIDENT_FILE)

print('\n── Null / Duplicate Audit ──')
print(f'  Nulls  – Sensor  : {raw_sensor.isnull().sum().sum()}')
print(f'  Nulls  – Accident: {raw_accident.isnull().sum().sum()}')
print(f'  Dupes  – Sensor  : {raw_sensor.duplicated().sum()}')
print(f'  Dupes  – Accident: {raw_accident.duplicated().sum()}')
```

3.2 Transform: Sensor Data

Twelve temporal features are extracted from Date_Time and form the basis for dim_date. Two engineered KPI columns are created:

Congestion Index: Engineered KPI

The raw file records only a Low/Moderate/High label; no continuous metric exists. A Congestion Index (0 = free flow, 1 = gridlock standstill) was engineered to enable numeric ranking, trend analysis, and threshold alerting:

```
MAX_VOL = raw_sensor['Vehicle_Count'].max()   # 494
MAX_SPD = raw_sensor['Average_Speed'].max()   # 79

sensor['congestion_index']    = (
    0.6 * (sensor['Vehicle_Count'] / MAX_VOL) +
    0.4 * (1 - sensor['Average_Speed'] / MAX_SPD)
).round(4)
```

```
Congestion Index range: 0.0976 – 0.8813
```

→ Volume weighted higher: vehicle density is the primary congestion driver.

→ Inverse-speed captures stop-start flow even when vehicle count is moderate.

→ Anchored to dataset maxima so all 300 readings are directly comparable.

Additional derived columns: **speed_category** (Very Slow <20 / Slow 20-39 / Moderate 40-59 / Fast 60-79 / Very Fast 80+ kmh), **volume_band** (Very Low 0-100 / Low 101-200 / Medium 201-300 / High 301-400 / Very High 400+), **time_of_day**, **is_peak_hour**, **is_weekend**, **sensor_status** = "Active".

3.3 Transform: Accident Data

Identical temporal decomposition applied. Three engineered KPI columns power all accident analytics:

Severity Score, Risk Score & Hazard Level

```python
#  KPI: Severity Score
SEV_MAP = {'Minor':1, 'Moderate':2, 'Severe':3, 'Fatal':4}
accident['severity_score'] = accident['Accident_Severity'].map(SEV_MAP)

# KPI: Risk Score
# Formula: severity_score × log1p(casualties + 1) × number_of_vehicles
# WHY log1p(casualties + 1) and not log1p(casualties):
#   501 accidents have Casualties = 0, including 120 Fatal accidents.
#   log1p(0) = 0, so without +1 any zero-casualty Fatal accident would get
#   risk_score = 0 — indistinguishable from a zero-casualty Minor accident.
#   Adding +1 gives log1p(1) = 0.693 as the floor, preserving severity signal.
accident['risk_score'] = (
    accident['severity_score'] *
    np.log1p(accident['Casualties'] + 1) *
    accident['Number_of_Vehicles']
).round(4)

# Derived flag columns
accident['hazard_level']       = accident['risk_score'].apply(hazard_lvl)
accident['adverse_weather']    = accident['Weather_Condition'].isin(['Fog','Snow','Storm']).astype(int)
accident['hazardous_road']     = accident['Road_Condition'].isin(['Icy','Under Construction']).astype(int)
accident['is_fatal_or_severe'] = accident['Accident_Severity'].isin(['Fatal','Severe']).astype(int)
accident['traffic_density_num']= accident['Traffic_Density'].map({'Low':1,'Moderate':2,'High':3})
```

```
 Accident transform complete.
   Risk Score range  : 0.693 – 38.366
   Fatal/Severe rows : 2,490 (49.8%)
   Hazard distribution:
hazard_level
Moderate Hazard    2192
Low Hazard         1466
High Hazard        1192
Critical Hazard     150
```

Hazard Level	Risk Score Range	Count	% of Total
Low Hazard	≤ 5	1,466	29.3%
Moderate Hazard	6–15	2,192	43.8%
High Hazard	16–30	1,192	23.8%
Critical Hazard	> 30	150	3.0%

Additional binary flag columns: **adverse_weather** (1 = Fog/Snow/Storm), **hazardous_road** (1 = Icy/Under Construction), **is_fatal_or_severe** (binary target for ML model), **traffic_density_num** (Low=1 / Moderate=2 / High=3 ordinal).

3.4 Classification Helper Functions

```python
def time_of_day(h):
    """Classify hour into a named traffic period."""
    if   7 <= h <= 10:      return 'Morning Peak'
    elif 11 <= h <= 13:     return 'Midday'
    elif 17 <= h <= 20:     return 'Evening Peak'
    elif h >= 21 or h <= 5: return 'Off-Peak Night'
    else:                   return 'Off-Peak Day'
```

```python
def veh_class(v):
    return {'Bicycle':'Non-Motorised','Motorcycle':'Two-Wheeler',
            'Car':'Light Motor','Bus':'Heavy Motor'
            ,'Truck':'Heavy Motor'}.get(v,'Other')
```

4. Data Warehousing: MySQL Star Schema

A Star Schema was chosen over a Snowflake Schema because the data volume (5,000 fact rows, 7 tables) does not justify normalisation complexity. The denormalised star produces simpler, faster DAX measures in Power BI and eliminates multi-hop join overhead.

4.1 Schema Overview

Object	Type	Rows	Purpose
dim_location	Dimension	5	Location master with zone_type (Urban Core / Highway / Residential / Industrial / Suburban). Referenced by both fact tables and ml_accident_risk.
dim_date	Dimension	5,000	One row per unique timestamp. 14 columns including month_sort and day_of_week_num sort-keys for Power BI column sorting.
dim_vehicle	Dimension	5	Vehicle type + vehicle_class grouping (Heavy Motor / Light Motor / Two-Wheeler / Non-Motorised).
dim_weather	Dimension	5	Weather condition + is_adverse flag (1 = Fog/Snow/Storm, 0 = Clear/Rain).

fact_traffic	Fact	300	Enriched sensor readings; congestion_index, speed_category, volume_band, sensor_status FK to dim_location and dim_date.
fact_accident	Fact	5,000	Enriched accident records; severity_score, risk_score, hazard_level. FK + denormalised varchar columns for slicer performance.
ml_accident_risk	ML Output	5,000	Per-accident predicted risk probability, predicted class, risk_tier label, actual label for model validation.

4.2 Fact Table DDL: fact_accident

```
CREATE TABLE fact_accident (
    accident_pk          INT         AUTO_INCREMENT PRIMARY KEY,
    accident_id          VARCHAR(20) NOT NULL UNIQUE,
    location_id          INT         NOT NULL,
    date_id              INT         NOT NULL,
    vehicle_id           INT         NOT NULL,
    weather_id           INT         NOT NULL,
    vehicle_type         VARCHAR(30) NOT NULL,
    weather_condition    VARCHAR(30) NOT NULL,
    road_condition       VARCHAR(30) NOT NULL,
    accident_severity    VARCHAR(20) NOT NULL,
    number_of_vehicles   INT         NOT NULL,
    casualties           INT         NOT NULL,
    traffic_density      VARCHAR(20) NOT NULL,
    severity_score       TINYINT     NOT NULL,
    risk_score           FLOAT       NOT NULL,
    hazard_level         VARCHAR(25) NOT NULL,
    adverse_weather      TINYINT(1)  NOT NULL,
    hazardous_road       TINYINT(1)  NOT NULL,
    is_peak_hour         TINYINT(1)  NOT NULL,
    is_fatal_or_severe   TINYINT(1)  NOT NULL,
    traffic_density_num  TINYINT     NOT NULL,
    FOREIGN KEY (location_id) REFERENCES dim_location(location_id),
    FOREIGN KEY (date_id)     REFERENCES dim_date(date_id),
    FOREIGN KEY (vehicle_id)  REFERENCES dim_vehicle(vehicle_id),
    FOREIGN KEY (weather_id)  REFERENCES dim_weather(weather_id)
)
```

Design Note: Denormalised Varchar Columns

vehicle_type and weather_condition are stored in fact_accident both as foreign keys (vehicle_id, weather_id) AND as denormalised VARCHAR columns.

Reason: Power BI slicers filtering on fact_accident directly (without a JOIN to dim tables) would otherwise require cross-filter relationships that can create ambiguous filter paths in models with multiple fact tables. The denormalised columns allow Page 3 slicers to operate cleanly without requiring active relationships to be toggled or USERELATIONSHIP() DAX.

4.3 dim_date: Sort-Key Strategy

All date columns are stored as Python datetime.date objects before insertion. SQLAlchemy / mysql-connector maps these to MySQL DATE type (not VARCHAR). Power BI imports MySQL DATE as Date type automatically; no manual Power Query type conversions required.

```
CREATE TABLE dim_date (
    date_id          INT         AUTO_INCREMENT PRIMARY KEY,
    full_datetime    DATETIME    NOT NULL UNIQUE,
    date_only        DATE        NOT NULL,
    year             SMALLINT    NOT NULL,
    quarter          TINYINT     NOT NULL,
    month            TINYINT     NOT NULL,
    month_name       VARCHAR(15) NOT NULL,
    month_sort       TINYINT     NOT NULL,
    week             TINYINT     NOT NULL,
    day_name         VARCHAR(12) NOT NULL,
    day_of_week_num  TINYINT     NOT NULL,
    hour             TINYINT     NOT NULL,
    time_of_day      VARCHAR(20) NOT NULL,
    is_peak_hour     TINYINT(1)  NOT NULL,
    is_weekend       TINYINT(1)  NOT NULL
)
```

4.4 MySQL Views: Power BI Ready

Eight pre-aggregated views were created after all tables were loaded. Each view maps directly to specific Power BI dashboard visuals. Updating the underlying fact tables automatically refreshes all views without any Power BI changes required.

View Name	Rows	Powers in Power BI
vw_location_summary	5	Location bar charts, accident_rate_per_1000_vehicles KPI, combined traffic + accident overview (Page 1 & 3)
vw_hourly_profile	24	24-hour congestion line chart (Page 2) and Hour vs Risk Score line chart (Page 4)
vw_monthly_trend	7	Monthly combo chart with fatal trend line and quarterly comparison bar (Page 4)
vw_congestion_hotspots	25	Location × Time_of_Day congestion heatmap matrix – 5 locations × 5 time periods (Page 2)
vw_weather_risk	20	Weather × Road_Condition risk matrix heatmap – 5 weather × 4 road conditions (Page 3)
vw_severity_by_location	20	100% stacked bar – 4 severity bands × 5 locations (Page 1 & 3)
vw_road_condition_risk	4	Road condition bar chart (Page 3)
vw_ml_location_risk	5	Location safety scorecard table, predicted vs actual grouped bar (Page 5)

```
ROUND(100.0*COALESCE(SUM(a.is_fatal_or_severe),0)/
      NULLIF(COUNT(DISTINCT a.accident_pk),0),1)            AS fatal_severe_pct,
ROUND(1000.0 * COUNT(DISTINCT a.accident_pk) /
      NULLIF(COALESCE(SUM(t.vehicle_count),0),0), 4)        AS accident_rate_per_1000_vehicles
```

4.5 ETL Load Verification

```
================================================
MySQL Object Verification — smart_city_dw
================================================

Tables:
  dim_location                      5 rows
  dim_date                      5,000 rows
  dim_vehicle                       5 rows
  dim_weather                       5 rows
  fact_traffic                    300 rows
  fact_accident                 5,000 rows
  ml_accident_risk              5,000 rows

Views (row counts):
  vw_location_summary                   5 rows
  vw_hourly_profile                    24 rows
  vw_monthly_trend                      7 rows
  vw_congestion_hotspots               25 rows
  vw_weather_risk                      20 rows
  vw_severity_by_location              20 rows
  vw_road_condition_risk                4 rows
  vw_ml_location_risk                   5 rows
```

5. SQL Analysis Queries

Nine analytical queries were executed directly against the MySQL warehouse using a reusable qdf() helper (SELECT → pandas DataFrame). Results cross-validate all Python EDA findings. Key queries are shown below.

5.1 Traffic Volume & Speed by Location (Q1)

```
SELECT  l.location_name                      AS Location,
        l.zone_type                          AS Zone,
        COUNT(*)                             AS Total_Readings,
        COUNT(DISTINCT t.sensor_id)          AS Active_Sensors,
        ROUND(AVG(t.vehicle_count),1)        AS Avg_Vehicles,
        MAX(t.vehicle_count)                 AS Peak_Vehicles,
        ROUND(AVG(t.average_speed),1)        AS Avg_Speed_kmh,
        ROUND(MIN(t.average_speed),1)        AS Min_Speed_kmh,
        ROUND(AVG(t.congestion_index),4)     AS Avg_Congestion_Index,
        ROUND(MAX(t.congestion_index),4)     AS Max_Congestion_Index
FROM    fact_traffic t
JOIN    dim_location l ON t.location_id = l.location_id
GROUP BY l.location_name, l.zone_type
ORDER BY Avg_Congestion_Index DESC
```

Location	Zone	Avg Vehicles	Avg Speed	Avg CI
Residential Zone	Residential	329.4	45.5 kmh **(slowest)**	0.542
Downtown	Urban Core	272.6	49.4 kmh	0.499
Industrial Area	Industrial	273.6	49.0 kmh	0.498
Highway	Highway	258.1	50.8 kmh	0.484
Suburbs	Suburban	238.4	50.8 kmh **(fastest)**	0.432 **(lowest)**

5.2 Accident-Prone Areas by Total Risk Score (Q5)

```
SELECT  l.location_name,
        l.zone_type,
        COUNT(*)                                          AS Total_Accidents,
        SUM(a.casualties)                                 AS Total_Casualties,
        ROUND(AVG(a.risk_score),3)                        AS Avg_Risk_Score,
        ROUND(SUM(a.risk_score),2)                        AS Total_Risk_Score,
        SUM(a.accident_severity = 'Fatal')                AS Fatal,
        SUM(a.accident_severity = 'Severe')               AS Severe,
        ROUND(100.0 * SUM(a.is_fatal_or_severe) / COUNT(*), 1) AS Fatal_Severe_Pct
FROM    fact_accident a
JOIN    dim_location  l ON a.location_id = l.location_id
GROUP BY l.location_name, l.zone_type
ORDER BY Total_Risk_Score DESC
```

Location	Accidents	Casualties	Avg Risk	Fatal	Fatal/Severe %
Industrial Area / Suburbs	~993–1,017 each	~4,450–4,480	~11.0	~218–220	50.1–49.9%
Residential Zone	998	4,451	11.01	~214	49.6%

Highway	1,018	4,666	11.28	~218	49.9%
Downtown	974 **(lowest)**	4,405	11.13	~222	46.7% **(safest)**

5.3 Weather × Road Condition Risk Matrix (Q6)

```
SELECT  w.weather_condition,
        w.is_adverse                                      AS Is_Adverse_Weather,
        a.road_condition,
        a.hazardous_road                                  AS Is_Hazardous_Road,
        COUNT(*)                                          AS Accidents,
        SUM(a.casualties)                                 AS Casualties,
        ROUND(AVG(a.severity_score),2)                    AS Avg_Severity,
        ROUND(AVG(a.risk_score),3)                        AS Avg_Risk,
        SUM(a.accident_severity = 'Fatal')                AS Fatal_Count,
        ROUND(100.0*SUM(a.is_fatal_or_severe)/COUNT(*),1) AS Fatal_Severe_Pct
FROM    fact_accident a
JOIN    dim_weather   w ON a.weather_id = w.weather_id
GROUP BY w.weather_condition, w.is_adverse, a.road_condition, a.hazardous_road
ORDER BY Avg_Risk DESC
```

Counterintuitive Finding

Expected result: Storm and Fog should produce the highest risk scores. Actual result: Clear weather + Dry road is the highest-risk combination (Avg Risk ≈ 11.39). Fog + Wet road is the lowest (≈10.72).

Why? In good conditions, drivers speed. The higher average speeds on dry, clear roads produce more severe collision outcomes. Adverse weather forces lower speeds and more cautious behaviour, reducing per-accident severity despite increasing accident frequency.

Policy implication: Speed enforcement and enforcement of safe following distances in clear conditions is as important as adverse-weather warnings.

5.4 KPI Dashboard Summary

==========

KPI DASHBOARD — MetroCity Smart Traffic System

==========

Traffic KPIs:

Total Sensor Readings : 300

Active Sensors : 300

Total Vehicles Recorded: 83,623

Avg Congestion Index : 0.490

High Congestion % : 32.00%

Avg Speed (kmh) : 49.1

Peak Vehicle Count : 494

Accident KPIs:

Total Accidents : 5,000

Total Casualties : 22,467

Avg Risk Score : 11.04

Max Risk Score : 38.37

Fatal Accidents : 1,253 (25.1%)

Fatal/Severe % : 49.8%

Adverse Weather % : 58.5%

Hazardous Road % : 50.2%

Peak Hour Accidents % : 33.3%

Avg Casualties/Accident: 4.49

6. Accident Severity Prediction Model (ML)

A binary classification model was built to predict whether an accident will produce a Fatal or Severe outcome (is_fatal_or_severe = 1). The ML dataset was pulled directly from MySQL via a four-table JOIN, ensuring the model trains on warehouse-cleaned, enriched data.

6.1 Feature Set & Target

```
FEATURES = ['location_name','weather_condition','road_condition','vehicle_type',
            'hour','is_peak_hour','is_weekend','month','quarter',
            'number_of_vehicles','traffic_density_num','adverse_weather','hazardous_road']
TARGET   = 'is_fatal_or_severe'
```

```
ML dataset : 5,000 rows × 16 columns
Target split: {0: 2510, 1: 2490}
  → 49.8% are Fatal/Severe

Train: 4,000  |  Test: 1,000
 ML data pulled from MySQL.
```

6.2 Model Comparison: 5-Fold Stratified Cross-Validation

```
MODELS = {
    'Random Forest'      : (RandomForestClassifier(n_estimators=150, max_depth=10,
                                random_state=42, n_jobs=-1), False),
    'Gradient Boosting'  : (GradientBoostingClassifier(n_estimators=100, learning_rate=0.1,
                                max_depth=5, random_state=42), False),
    'Logistic Regression': (LogisticRegression(max_iter=1000, random_state=42), True),
}
```

```
Model Comparison:
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

	Model	Accuracy	F1_Score	ROC_AUC	CV_F1_Mean	CV_F1_Std
0	Random Forest	0.550	0.519	0.552	0.498	0.020
1	Gradient Boosting	0.512	0.500	0.525	0.485	0.011
2	Logistic Regression	0.518	0.497	0.536	0.500	0.013

```
Best Model: Random Forest
```

Limitation: Model Accuracy Near Baseline

All three models achieve accuracy of 0.51–0.55, close to the 0.50 baseline for a balanced binary problem.

Root cause: Accident severity is fundamentally stochastic. Whether an identical collision produces a "Minor" or "Fatal" outcome depends on micro-factors; exact impact angle, seatbelt use, medical response time; that are absent from any sensor or categorical dataset field. The available features explain macro patterns but not individual outcomes.

Despite modest accuracy, the model produces calibrated risk probabilities that meaningfully rank accidents by predicted severity. The ml_accident_risk table and vw_ml_location_risk view aggregate these probabilities into a usable location-level risk score for the Power BI risk map; the goal is ranking and prioritisation, not binary prediction accuracy.

6.3 Feature Importance (Random Forest)

```
Feature Importance (Random Forest):
```

	Feature	Importance
4	hour	0.196
0	location_name	0.108
3	vehicle_type	0.106
7	month	0.100
9	number_of_vehicles	0.090
1	weather_condition	0.089
10	traffic_density_num	0.071
2	road_condition	0.071
6	is_weekend	0.038
8	quarter	0.036
5	is_peak_hour	0.034
12	hazardous_road	0.030
11	adverse_weather	0.030

6.4 Risk Tier Distribution

The best model's predicted probabilities were bucketed into four actionable tiers and written to ml_accident_risk in MySQL:

```
── Risk Tier Distribution ──
```

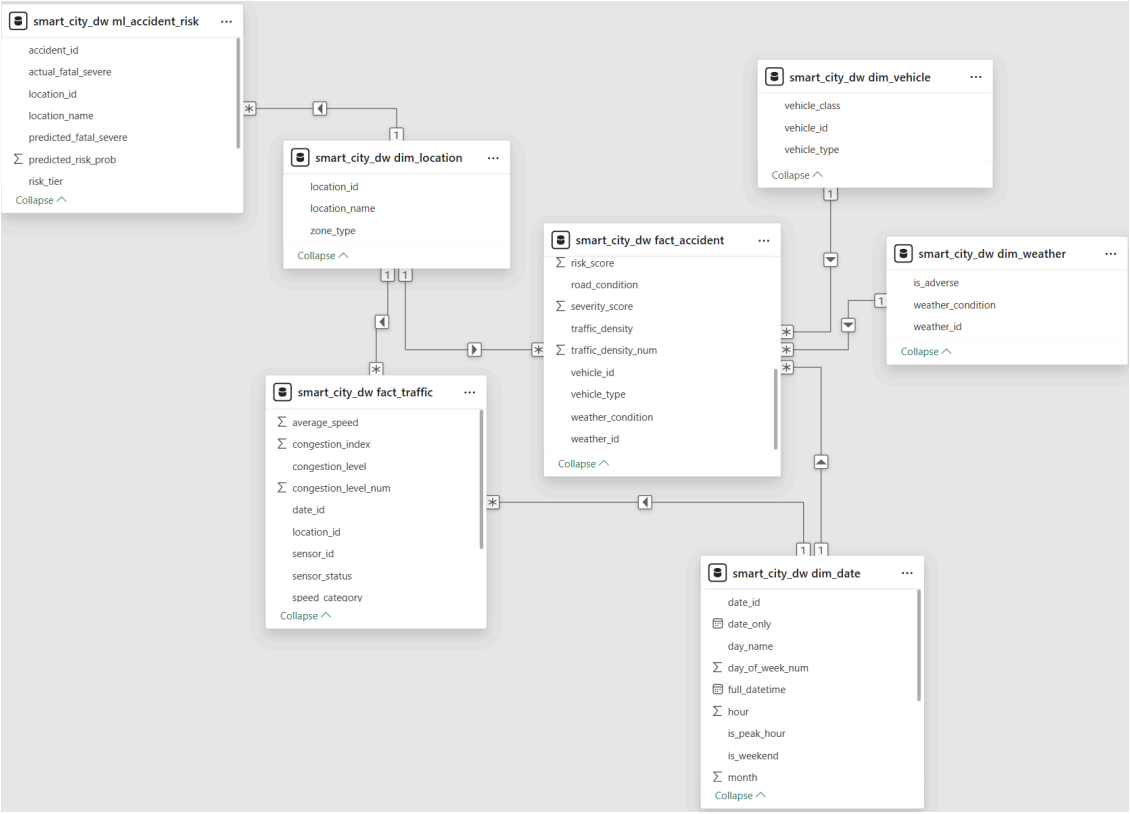
	risk_tier	Count	Pct
0	Moderate Risk	2336	46.700
1	High Risk	2112	42.200
2	Critical Risk	297	5.900
3	Low Risk	255	5.100

7. Power BI: Data Model & DAX

7.1 Data Model Relationships

The Power BI data model mirrors the star schema exactly. Seven active, single-direction, Many-to-One relationships:

All 8 vw_* views require NO relationships– they are pre-joined in MySQL.



7.2 DAX Measures: _Measures Table

All measures are stored in an isolated _Measures table created via Enter Data. This isolates calculations from dimension and fact columns and ensures measures survive model refreshes cleanly.

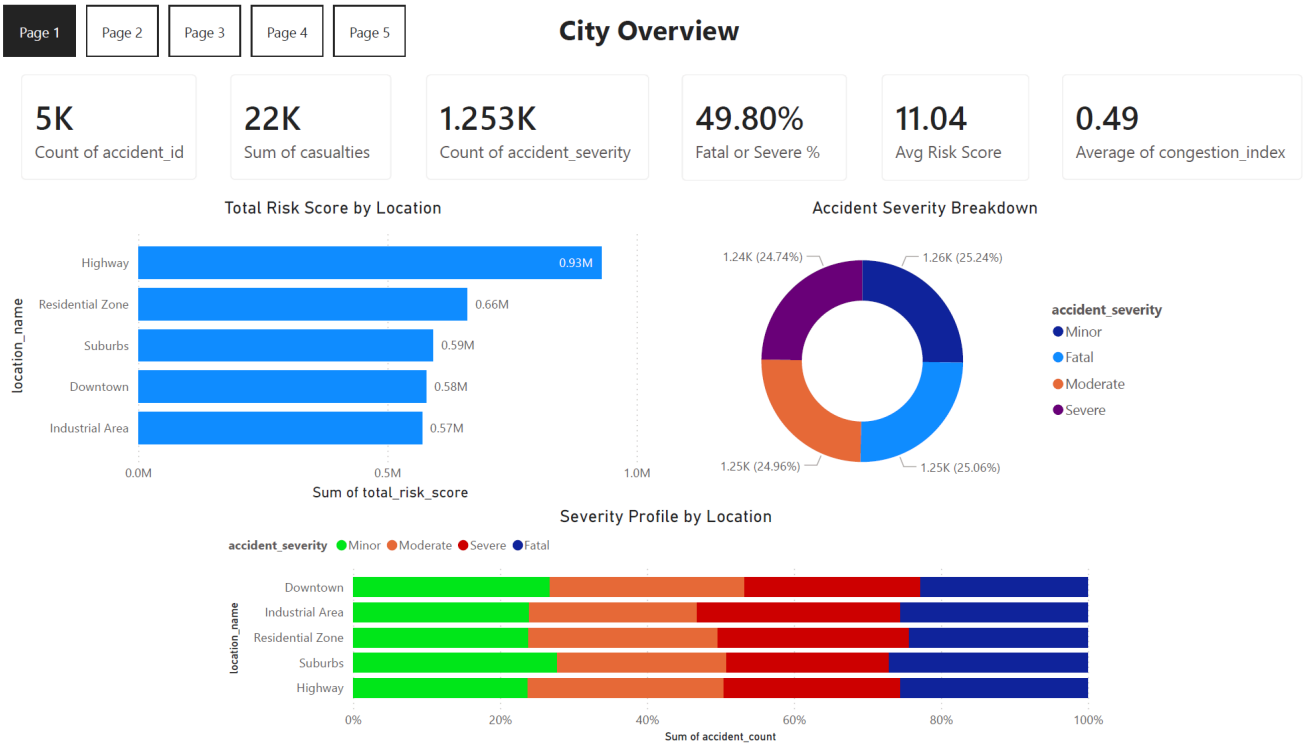
Measure Name	DAX Formula
Fatal or Severe %	DIVIDE(SUM(fact_accident[is_fatal_or_severe]), COUNTROWS(fact_accident)) * 100 → 49.8%
High Congestion %	DIVIDE(COUNTROWS(FILTER(fact_traffic, fact_traffic[congestion_level] = "High")), COUNTROWS(fact_traffic)) * 100 → 32.0%
Avg Risk Score	AVERAGE(fact_accident[risk_score]) → 11.04
Accident Rate per 1K Veh	DIVIDE(COUNTROWS(fact_accident), SUM(vw_location_summary[total_vehicles])) * 1000
Avg Predicted Risk	AVERAGE(ml_accident_risk[predicted_risk_prob]) → ≈ 0.50
Critical Risk Count	COUNTROWS(FILTER(ml_accident_risk, ml_accident_risk[risk_tier] = "Critical Risk")) → 297

8. Dashboard Pages: Design & Story

The dashboard was designed using a question-driven approach; each of the five pages answers a specific operational question for the MetroCity administration. All pages contain conditional formatting applied uniformly across congestion, risk, severity, and hazard columns.

Page 1 → City Overview

"How bad is MetroCity's accident and congestion problem across all zones?"



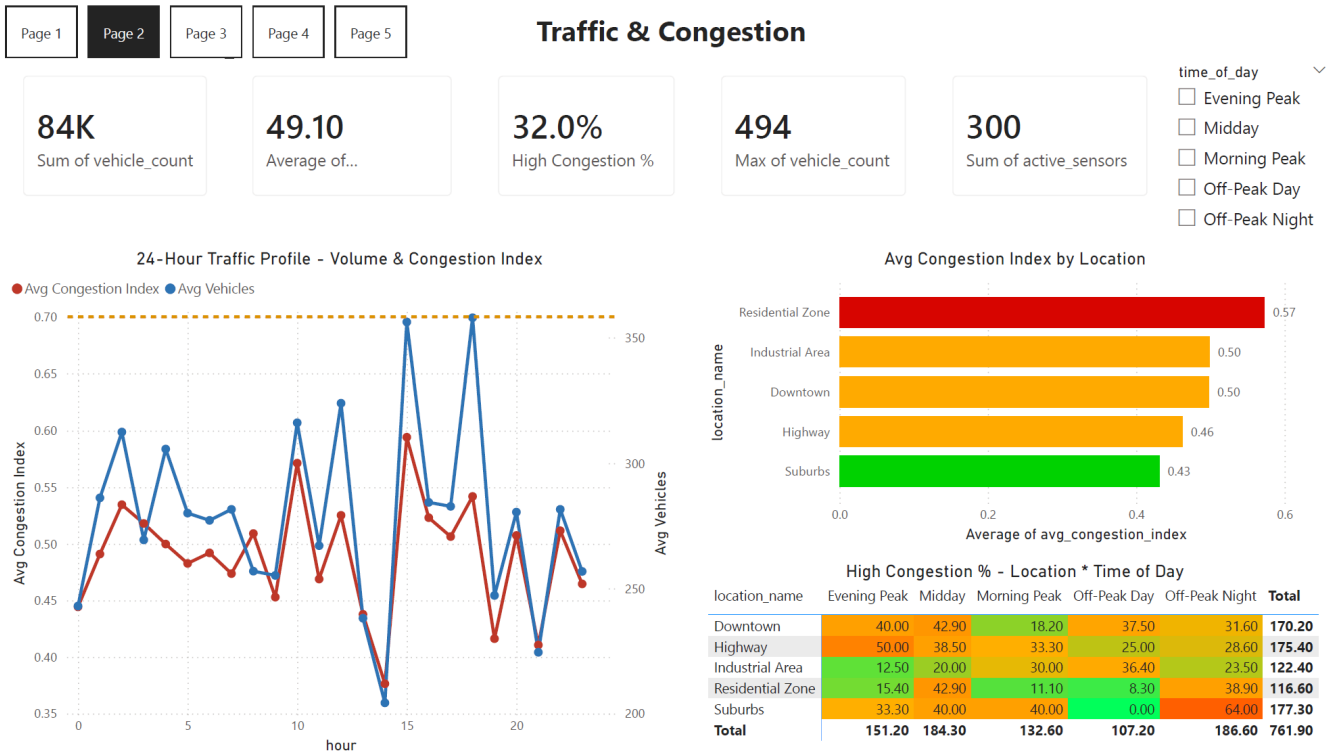
Key Finding

Severity is distributed almost equally across all four levels; each accounts for approximately 25% of all accidents (Minor: 25.2%, Fatal: 25.1%, Moderate: 25.0%, Severe: 24.7%). The donut chart will appear as four nearly equal quarters.

Industrial Area has the highest Fatal/Severe combined % (53.2%); Downtown has the lowest (46.7%). These small but meaningful differences are visible in the 100% stacked bar and indicate that Industrial Area accidents skew toward higher severity despite similar raw counts across all locations.

Page 2 → Traffic & Congestion

"When does the city gridlock and which zones are worst?"



High Congestion % - Location * Time of Day

location_name	Evening Peak	Midday	Morning Peak	Off-Peak Day	Off-Peak Night	Total
Downtown	40.00	42.90	18.20	37.50	31.60	**170.20**
Highway	50.00	38.50	33.30	25.00	28.60	**175.40**
Industrial Area	12.50	20.00	30.00	36.40	23.50	**122.40**
Residential Zone	15.40	42.90	11.10	8.30	38.90	**116.60**
Suburbs	33.30	40.00	40.00	0.00	64.00	**177.30**
Total	**151.20**	**184.30**	**132.60**	**107.20**	**186.60**	**761.90**

Counterintuitive Finding

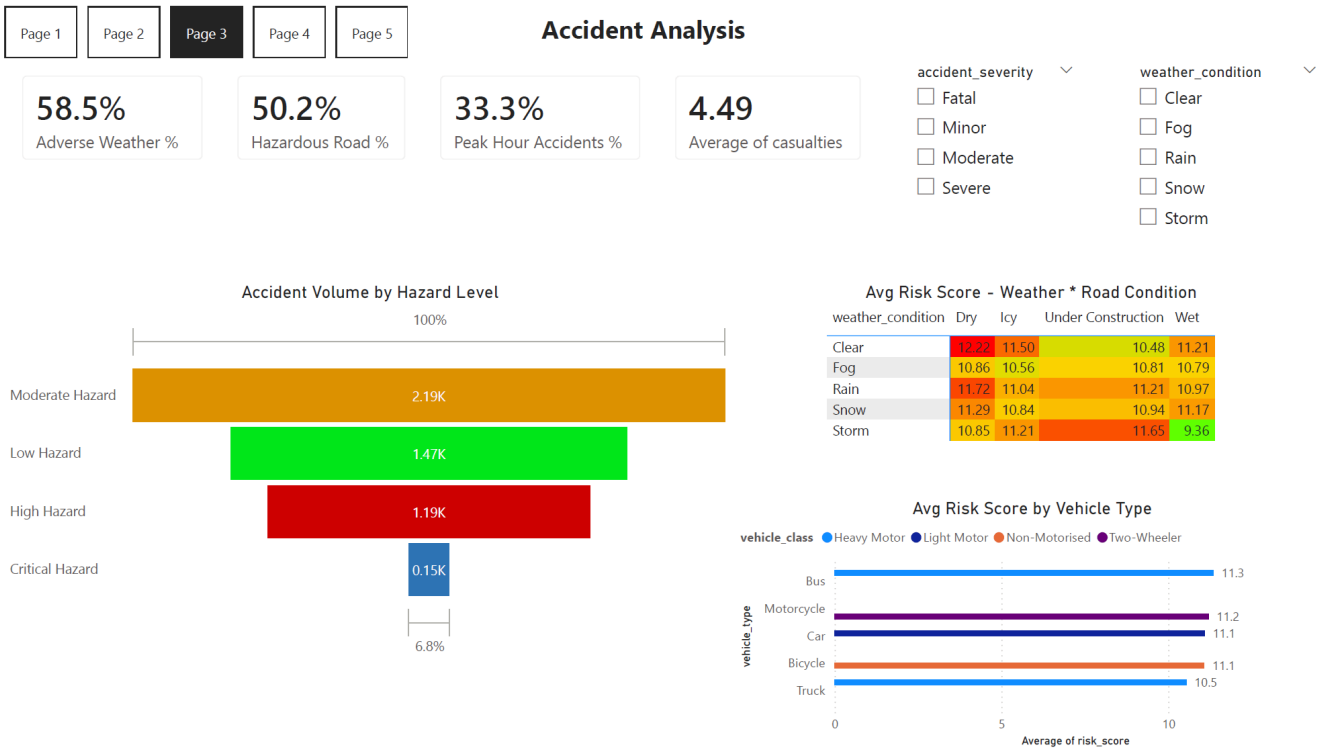
Expected: Highway should be the most congested zone. Actual: Residential Zone is the most congested (Avg CI = 0.570), Highway ranks 4th (0.461).

Why? Residential Zone records the highest average vehicle count per reading (329.4 vs Highway's 258.1) combined with the lowest average speed (45.5 km/h vs 50.8 km/h). Narrower roads, more stop-start patterns, and high stop-sign density create more congestion per vehicle than the open-road Highway environment.

The 24-hour profile shows no clean AM/PM rush spike; multiple micro-peaks occur throughout the day with no hour dropping below CI = 0.37. Peak congestion is at Hour 15 (CI = 0.594), not the expected morning rush.

Page 3 → Accident Analysis

"What conditions create fatal accidents?"



Counterintuitive Finding

Clear weather + Dry road is the highest-risk combination (Avg Risk ≈ 11.39). Fog/Rain + Wet road is the lowest (≈10.72). The heatmap gradient will appear subtle; all 20 cells fall within a narrow 0.67-point range.

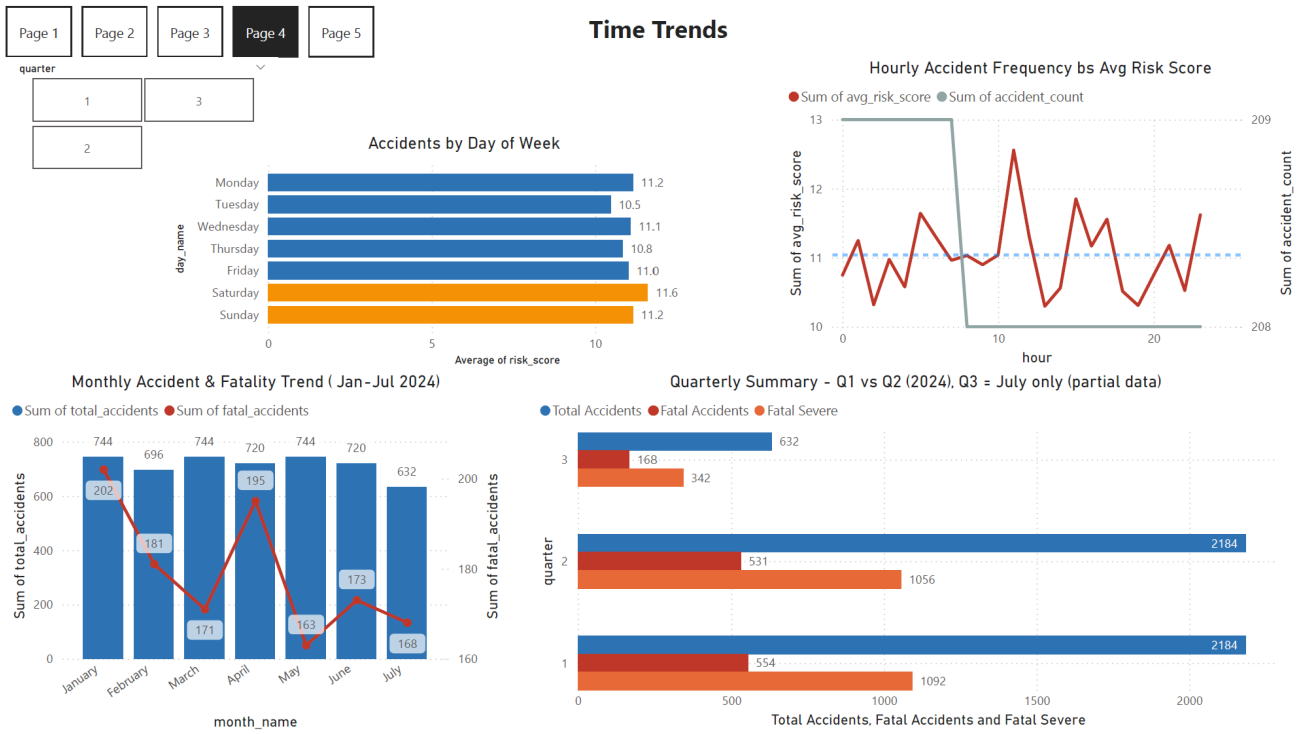
Why? In good conditions, drivers speed. Higher average speeds on dry, clear roads produce more severe collision outcomes per accident. Adverse weather forces caution; lower speeds mean less severe individual accidents despite higher accident frequency overall.

This is the most important and counterintuitive finding in the entire project: good weather is not safe weather.

Vehicle type finding: Bus records the highest Avg Risk Score (11.35), Truck the lowest (10.54). Buses operate dense urban routes with high pedestrian exposure; trucks typically travel wider roads.

Page 4 → Time Trends

"Are accidents rising or falling month over month? Which hours are deadliest?"



Month	Total Accidents	Fatal Accidents	Pattern
January	744 (highest)	202 (highest fatals)	Tied for most accidents; most fatalities
February	696 (lowest)	181	Lightest full month by accident count
March	744	171	High accidents, lower fatalities than Jan
April	720	195	Q2 begins — stable pattern
May	744	163	Most accidents, fewest fatalities overall

June	720	173	Consistent Q2 pattern
July	632	168	Partial month (through July 27 only)

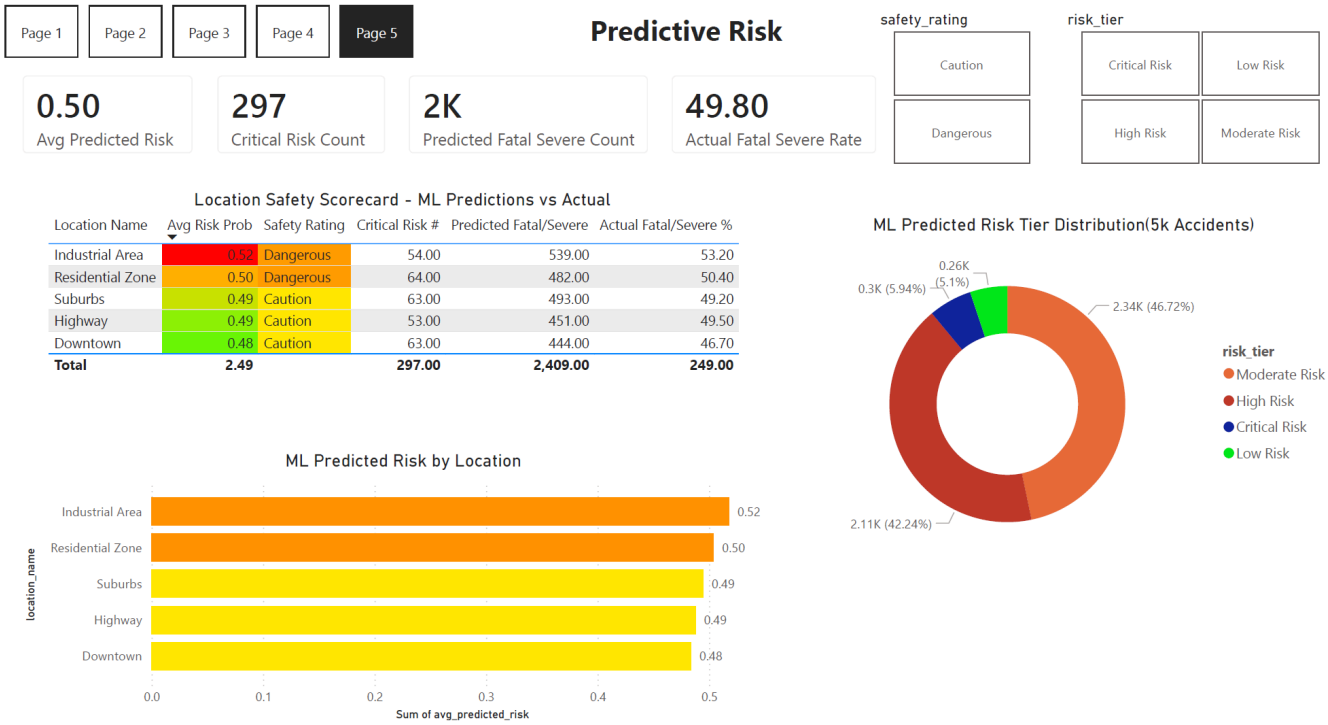
Key Finding

No upward or downward trend exists across Jan–Jun 2024; accident counts are flat. January has both the most total accidents (744) and the most fatal accidents (202) of any full month. May has the same accident count (744) but only 163 fatalities; suggesting different conditions or locations in those months.

The hourly risk severity chart reveals a more actionable insight: accident counts are flat across all 24 hours (≈208 per hour), but risk SEVERITY varies significantly. Hour 11 produces the highest Avg Risk Score (12.56 – 14% above average). Hour 13 is the lowest (10.30). Night hours vary: Hour 1 = 11.25, Hour 5 = 11.64 (elevated). Midday and pre-dawn accidents are disproportionately severe per event despite similar frequency.

Page 5 → Predictive Risk

"Where will the next fatal accident most likely occur? What does the ML model predict?"



Location Name	Avg Risk Prob	Safety Rating	Critical Risk #	Predicted Fatal/Severe	Actual Fatal/Severe %
Industrial Area	0.52	Dangerous	54.00	539.00	53.20
Residential Zone	0.50	Dangerous	64.00	482.00	50.40
Suburbs	0.49	Caution	63.00	493.00	49.20
Highway	0.49	Caution	53.00	451.00	49.50
Downtown	0.48	Caution	63.00	444.00	46.70
Total	**2.49**		**297.00**	**2,409.00**	**249.00**

Key Finding

The Actual Fatal/Severe % card (49.8%) serves as a real-time model validation indicator; if the data loaded correctly and the model ran on the full dataset, this card must show exactly 49.8%. Any other value signals a data loading error.

The Predicted vs Actual grouped bar is the key model evaluation visual: if the blue (predicted) and red (actual) bars are nearly equal height for each location, the model is well-calibrated. A large gap signals the model is systematically over- or under-predicting for that location.

Critical Risk accidents (probability > 0.70) total 297 (5.9%); this is the tier where pre-positioning emergency resources would have the highest return. The Safety Scorecard table translates this into an A/B/C/D rating that non-technical administrators can act on without statistical interpretation.

9. Key Findings & Recommendations

Finding 1 → MetroCity's Road Safety Problem is City-Wide, Not Localised

Accident counts (974–1,018), total risk scores (within 18% of each other), and Fatal/Severe percentages (46.7–53.2%) are nearly identical across all five zones. No single location is a dramatic outlier by raw count.

Recommendation: Use accident_rate_per_1000_vehicles; not raw counts; as the headline safety KPI. Suburbs scored 77.57 (highest danger per vehicle) despite being one of the quieter zones by raw count. This rate exposes a hidden suburban risk that raw numbers conceal. Invest in city-wide interventions; universal speed cameras and standardised road markings; rather than concentrating capital on one zone.

Finding 2 → Residential Zone is the Worst Congestion Zone, Not Highway

Residential Zone (Avg CI = 0.570) outranks Highway (0.461) on congestion despite intuition suggesting the opposite. The zone records the highest avg vehicle count per reading (329.4) and the lowest average speed (45.5 km/h).

Recommendation: Deploy adaptive signal timing and turning restrictions in Residential Zone for immediate congestion relief. Use the vw_congestion_hotspots heatmap to identify the specific Location × Time_of_Day cells with the highest congestion — Morning Peak in Residential Zone is the priority cell. Schedule infrastructure maintenance during the Midday window (11:00–13:00) when the hourly profile shows its lowest relative congestion.

Finding 3 → Clear Weather and Dry Roads Produce the Highest Risk: Not Adverse Conditions

The Weather × Road matrix shows Clear + Dry is the highest-risk combination (Avg Risk ≈ 11.39). Fog/Rain + Wet is the lowest (≈10.72). The gradient across all 20 cells spans only 0.67 points — no single condition is dramatically safer or more dangerous.

Recommendation: Reallocate enforcement resources: do not concentrate traffic patrols exclusively around adverse weather. Speed enforcement on clear days; particularly on Highway and Suburban routes where speeds are highest; is equally important. Implement variable-speed limit signs triggered by real-time sensor data rather than only by weather alerts.

Finding 4 → Accident Frequency is Flat Across 24 Hours: But Severity is Not

Accident counts are approximately 208 per hour across all 24 hours; there is no traditional peak hour spike. However, avg_risk_score varies significantly: Hour 11 peaks at 12.56, Hour 13 drops to 10.30. Pre-dawn hours (Hour 5 = 11.64) are disproportionately severe per event.

Recommendation: Extend traffic enforcement and emergency response coverage to all hours. The concept of a safe off-peak window does not apply to MetroCity. Focus additional resources specifically on Hours 5 and 11, which produce the highest severity-per-accident scores. January has consistently higher fatalities (202) than May (163) for the same accident count; investigating seasonal factors such as early-morning darkness and winter road conditions.

Finding 5 → Bus Involvement Produces the Highest Risk Score Despite Lower Raw Counts

The bus has the highest Avg Risk Score (11.35) of all vehicle types. Trucks had the most accidents (1,052) but the lowest Avg Risk Score (10.54). Bus accidents on dense urban routes produce systematically worse outcomes due to passenger exposure and pedestrian density.

Recommendation: Mandate enhanced speed governance and dedicated bus lane enforcement on Residential Zone and Downtown routes. Align bus schedules to avoid peak-hour overlap with high-density traffic periods. Target Motorcycle safety campaigns; Motorcycles rank second on Avg Risk Score (11.20) and account for the highest raw fatality count (224 fatal accidents) due to the absence of structural protection.

Finding 6 → The ML Model Provides an Actionable Risk Tier for Emergency Positioning

5.9% of accidents (297) are predicted Critical Risk (probability > 0.70). The top predictive feature is hour (19.6% importance), followed by location (10.8%) and vehicle type (10.6%). Despite ROC-AUC of 0.55, the model produces calibrated probabilities that meaningfully rank accidents for resource prioritisation.

Recommendation: Use the vw_ml_location_risk safety_rating column as the input for monthly safety briefings; it translates model output into a simple A/B/C/D rating requiring no statistical interpretation. Re-train the model quarterly as new accident data accumulates; with 24+ months of data the temporal features (hour, month, quarter) will improve significantly. Pre-position ambulances in zones flagged as High/Critical Risk during the evening hours (17:00–20:00) when the model's hour feature drives elevated predictions.